

## Role

Act as a Senior Marketing & Sales Data Analyst, Customer Analytics Consultant, Business Intelligence (BI) Consultant, and Dashboard Designer

Your objective is to analyze the entire workbook as a real-world Marketing, Sales, and Customer Analytics dataset and transform the raw data into actionable business insights

Study the entire workbook before starting. Analyze all worksheets, columns, relationships, data types, calculations, and business logic

Do not make assumptions that are not supported by the data

Use only evidence from the dataset and clearly distinguish between

- Observed facts \*
- Calculated metrics \*
- Derived insights \*
- Business recommendations \*

Keep the original worksheets and raw data completely unchanged. Create all analysis, calculations, reports, and visualizations in new worksheets

---

## Data Quality & Validation Report .1

Create a comprehensive Data Quality Report for the entire workbook

Analyze

- Missing values \*
- Duplicate records \*
- Duplicate customers \*
- Duplicate transactions \*
- Invalid or inconsistent dates \*
- Incorrect date sequences \*
- Inconsistent categorical values \*
- Formatting inconsistencies \*
- Invalid numerical values \*
- Negative or impossible values \*
- Outliers \*
- Zero values where they may indicate data problems \*
- Inconsistent customer information \*
- Inconsistent campaign information \*
- Invalid sales or revenue values \*
- Logical inconsistencies between marketing and sales fields \*
- Potential data leakage \*

Columns with suspicious or unreliable values \*

:For marketing and customer datasets, specifically validate logical relationships such as

Sign-Up Date vs. Last Purchase Date \*

Campaign Date vs. Customer Activity \*

Email Sent vs. Email Open \*

Email Open vs. Email Click \*

Email Click vs. Conversion \*

Website Visit vs. Purchase \*

Discount vs. Revenue \*

Quantity vs. Sales \*

Price vs. Revenue \*

Cost vs. Profit \*

Customer status vs. purchase behavior \*

.Identify every logical inconsistency with actual counts and percentages

:For every issue, provide

Issue \*

(Affected column(s) \*

Number of affected records \*

Percentage of dataset affected \*

Severity \*

Business impact \*

Recommended treatment \*

.Do not automatically delete problematic records unless explicitly required

:Clearly distinguish between

Data errors \*

Legitimate business behavior \*

Missing information \*

Potential anomalies \*

\_\_\_\_\_

## Marketing Performance Analysis .2

.Perform a complete Marketing Performance EDA

.Analyze the effectiveness of marketing activities and campaigns

:Evaluate

Campaign Performance

- Best-performing campaigns \*
- Worst-performing campaigns \*
- Campaign reach \*
- Engagement \*
- Open Rate \*
- (Click-Through Rate (CTR) \*
- Conversion Rate \*
- Purchase Rate \*
- Revenue generated \*
- Profit generated \*
- Cost per acquisition if data is available \*
- Return on Marketing Investment / ROI if required fields exist \*
- Customer acquisition performance \*

.Compare campaigns using actual data

:Identify

- ?Which campaigns generated the most customers \*
- ?Which campaigns generated the most revenue \*
- ?Which campaigns generated the highest conversion \*
- ?Which campaigns generated the highest engagement \*
- ?Which campaigns appear inefficient \*
- ?Which campaigns generated high engagement but low sales \*
- ?Which campaigns generated low engagement but high sales \*

.Never assume that high engagement means high business performance

\_\_\_\_\_

### Customer Analysis .3

.Perform a complete Customer Analytics Analysis

:Analyze customers based on

- Purchase behavior \*
- Revenue \*
- Profit \*
- Number of purchases \*
- Average Order Value \*
- Recency \*
- Frequency \*
- Monetary Value \*
- Engagement \*
- Campaign response \*
- Conversion behavior \*

Customer lifetime behavior if sufficient data exists \*

:Identify

Highest-value customers \*

Lowest-value customers \*

Most frequent customers \*

Most recent customers \*

Customers with high revenue but low frequency \*

Customers with high frequency but low value \*

Inactive customers \*

Potentially churned customers if supported by the data \*

Highly engaged but non-converting customers \*

Customers responding best to campaigns \*

Customers generating the highest profit \*

:If the data supports it, perform RFM Analysis

Recency \*

Frequency \*

Monetary Value \*

:Create meaningful customer segments such as

Champions \*

Loyal Customers \*

Potential Loyalists \*

New Customers \*

At Risk \*

Dormant / Inactive \*

High Value \*

Low Value \*

.Do not use predefined thresholds without explaining and validating them using the dataset

---

#### Sales & Revenue Analysis .4

.Perform a complete Sales Analysis

:Analyze

Total Sales / Revenue \*

Total Profit \*

Total Orders \*

Total Customers \*

Average Order Value \*

Average Revenue per Customer \*  
Profit Margin \*  
Quantity Sold \*  
Discount impact \*  
Product performance \*  
Category performance \*  
Customer performance \*  
Regional performance \*  
Campaign-attributed performance if available \*

:Identify

Best Performers

Best Products \*  
Best Categories \*  
Best Customers \*  
Best Regions \*  
Best Campaigns \*  
Best Customer Segments \*

Worst Performers

Underperforming Products \*  
Weak Categories \*  
Low-value Customers \*  
Weak Regions \*  
Low-performing Campaigns \*  
Low-converting segments \*

.Support every conclusion with actual calculated values

---

## Customer Journey & Funnel Analysis .5

.Build a Customer Marketing Funnel based only on fields available in the dataset

:Analyze the progression from

Customer Acquisition → Sign-Up → Email Sent → Email Open → Email Click → Website  
Engagement → Conversion → Purchase → Revenue

.Only include stages that actually exist in the workbook

:For every stage calculate

Number of customers \*

Conversion rate \*  
Drop-off rate \*  
Percentage of customers progressing to the next stage \*

.Identify the biggest funnel drop-offs

.For every significant drop-off perform Root Cause Analysis

:Investigate whether the drop-off is associated with

Campaign \*  
Customer segment \*  
Region \*  
Product \*  
Price \*  
Discount \*  
Channel \*  
Customer characteristics \*  
Engagement level \*  
Time period \*

.Never claim causation unless the dataset provides sufficient evidence

\_\_\_\_\_

## Time Intelligence & Trend Analysis .6

.Create a complete Time Intelligence Analysis

:Analyze

Daily trends if appropriate \*  
Weekly trends if appropriate \*  
Monthly trends \*  
Quarterly trends \*  
Yearly trends \*  
MoM \*  
QoQ \*  
YoY if multiple years exist \*

:Track

Revenue \*  
Profit \*  
Orders \*  
Customers \*  
Campaign activity \*  
Engagement \*

Conversions \*  
Conversion Rate \*

.Highlight significant increases and decreases

.Whenever a major change is detected, perform a Root Cause Analysis

:For every major change answer

?What happened

.Describe the change using actual numbers

?When did it happen

.Identify the exact period

?Where did it happen

.Identify affected regions, channels, campaigns, products, or customer segments

?Who contributed

.Identify customers or customer segments when relevant

?What contributed

:Investigate

Campaign changes \*  
Discount changes \*  
Price changes \*  
Product mix \*  
Customer mix \*  
Marketing engagement \*  
Conversion changes \*  
Regional performance \*  
Order volume \*  
Costs \*

?What should the business do

.Provide evidence-based recommendations

.Never guess the cause

\_\_\_\_\_

## Marketing Channel Analysis .7

.If marketing channels are available, analyze each channel

:Compare

- Reach \*
- Engagement \*
- Open Rate \*
- CTR \*
- Conversion Rate \*
- Customers Acquired \*
- Revenue \*
- Profit \*
- Marketing Cost \*
- ROI / ROAS if available \*

:Identify

- Most efficient channels \*
- Highest-revenue channels \*
- Highest-converting channels \*
- High-cost low-return channels \*
- High-engagement low-conversion channels \*
- Low-engagement high-conversion channels \*

Recommend where marketing resources should be increased, maintained, tested, or  
.reduced

\_\_\_\_\_

## Campaign-to-Sales Analysis .8

.Connect marketing activity with customer and sales outcomes

:Determine

- ?Which campaigns generated purchases \*
- ?Which campaigns generated the highest revenue \*
- ?Which campaigns generated the highest profit \*
- ?Which campaigns acquired the most customers \*
- ?Which campaigns generated repeat customers \*
- ?Which campaigns generated high engagement but weak commercial results \*

:Compare campaign performance across

- Customer segments \*
- Products \*



Categories \*  
Regions \*  
Time periods \*  
Discounts \*  
Customer value \*

Identify campaigns that appear successful from a marketing perspective but weak from a  
.business perspective

---

## Customer Segmentation .9

.Create meaningful customer segments using evidence from the dataset

:Possible segmentation dimensions include

Recency \*  
Frequency \*  
Monetary Value \*  
Engagement \*  
Conversion behavior \*  
Campaign response \*  
Purchase behavior \*  
Geography \*  
Product preference \*  
Customer value \*

:For every segment provide

Number of customers \*  
Percentage of customers \*  
Revenue \*  
Profit \*  
Average Order Value \*  
Purchase frequency \*  
Conversion rate \*  
Engagement rate \*  
Recommended marketing strategy \*

.Avoid creating segments that cannot be supported by the available data

---

## Business Questions & Management Analysis .10

Create a Business Analysis sheet containing the most important questions management  
.should ask

:Examples

- ?Which campaigns generate the highest revenue .1
- ?Which campaigns generate customers but fail to generate purchases .2
- ?Which customer segments generate the highest value .3
- ?Which customers are at risk of becoming inactive .4
- ?Where is the largest conversion drop-off .5
- ?Which marketing channel performs best .6
- ?Which products generate the highest profit .7
- ?Are discounts improving sales or reducing profitability .8
- ?Which regions are underperforming .9
- ?Are highly engaged customers actually converting .10
- ?Which campaigns should be scaled .11
- ?Which campaigns should be optimized or stopped .12
- ?Which customer segments should receive retention campaigns .13
- ?Where are the biggest revenue opportunities .14
- ?What are the biggest business risks .15

.Answer every question using actual dataset evidence

---

Root Cause Analysis .11

Whenever a major business problem, anomaly, sales decline, conversion decline, profit decline, or campaign underperformance is detected, automatically perform a Root Cause Analysis

:Use the following structure

Problem

?What exactly went wrong

Magnitude

?How large is the impact

Timing

?When did it occur

Location

?Which region/channel/campaign/product/customer segment was affected

Drivers

?Which measurable factors contributed

Evidence

.Show the actual values and comparisons

Business Impact

.Explain the potential impact on revenue, profit, customers, or marketing efficiency

Recommended Action

.Provide a specific evidence-based action

:Do not use generic explanations such as

"Customers lost interest" \*

"The market changed" \*

"Competition increased" \*

"The campaign was ineffective" \*

.unless the dataset contains evidence supporting those claims

\_\_\_\_\_

Business Risks & Opportunities . 12

:Identify the most important

Business Risks

:Examples

Revenue decline \*

Profit decline \*

High discount dependency \*

Low conversion \*

High campaign cost \*

Customer inactivity \*

Customer concentration \*

Weak regions \*

Underperforming products \*

High engagement but low conversion \*

Data quality problems \*

Business Opportunities

:Examples

- High-value customer segments \*
- High-performing campaigns \*
- High-converting channels \*
- Cross-selling opportunities \*
- Upselling opportunities \*
- Retention opportunities \*
- Underpenetrated regions \*
- High-margin products \*
- Customers with strong engagement but no purchase \*

:Rank opportunities and risks by

- Business impact \*
- Data evidence \*
- Priority \*

\_\_\_\_\_

Data Modeling .13

.Evaluate the entire workbook structure

:Determine whether the current structure is appropriate for

- Marketing analytics \*
- Customer analytics \*
- Sales analytics \*
- BI reporting \*

:Recommend whether tables should be

- Merged \*
- Split \*
- Normalized \*
- Restructured \*

.Design the most appropriate Star Schema or Snowflake Schema

:Identify

Fact Tables

:Examples where supported

- Fact Sales \*
- Fact Orders \*

Fact Marketing Campaigns \*  
Fact Customer Engagement \*  
Fact Conversions \*

Dimension Tables

:Examples

Dim Customer \*  
Dim Product \*  
Dim Campaign \*  
Dim Date \*  
Dim Region \*  
Dim Channel \*  
Dim Category \*

:For every table specify

Grain \*  
Primary Key \*  
Foreign Keys \*  
Important measures \*  
Important dimensions \*

.Explain why the proposed model is appropriate

\_\_\_\_\_

Executive Dashboard .14

.Create a clean, modern, management-ready Executive Marketing & Sales Dashboard

.The dashboard should include the most valuable KPIs

KPI Cards

:Where data is available

Total Revenue \*  
Total Profit \*  
Total Customers \*  
Total Orders \*  
Conversion Rate \*  
Average Order Value \*  
Customer Acquisition \*  
Marketing Engagement \*  
Campaign Performance \*  
Profit Margin \*

## Customer Retention / Repeat Purchase Rate \*

### Recommended Visuals

:Include appropriate

- Revenue Trend \*
- Profit Trend \*
- Marketing Funnel \*
- Campaign Performance \*
- Customer Segment Performance \*
- Revenue by Region \*
- Revenue by Product / Category \*
- Customer Value Distribution \*
- Conversion Rate by Campaign \*
- Revenue vs. Discount \*
- Engagement vs. Conversion \*
- Top and Bottom Performers \*

.Do not create charts simply to fill space

.Every visual must answer a meaningful business question

---

## Automatic Business Problem Detection .15

.The dashboard should automatically highlight important problems and opportunities

:Detect issues such as

- Sales decline \*
- Profit decline \*
- Conversion decline \*
- High discount \*
- Low campaign ROI \*
- High engagement but low conversion \*
- Underperforming campaign \*
- Underperforming product \*
- Underperforming region \*
- Customer inactivity \*
- Customer concentration \*
- Unusual revenue spikes \*
- Unusual revenue drops \*
- Data quality issues affecting KPIs \*

:When an important issue is detected, create a visual or insight card showing

Problem → Evidence → Root Cause → Business Impact → Recommended Action

Executive Notebook .16

.Create a presentation-ready Executive Notebook designed for management review

:Include

Executive Highlights

.Summarize the most important findings

Key KPIs

.Present the major business metrics

Major Marketing Insights

.Explain campaign, channel, engagement, and conversion performance

Major Customer Insights

.Explain customer value, segmentation, behavior, retention, and purchasing patterns

Sales Insights

.Explain revenue, profit, products, categories, regions, and customer contribution

Root Causes

.Explain the evidence behind major business changes

Business Problems

.List the most important problems discovered

Opportunities

.Identify the highest-value growth opportunities

Recommendations

.Provide prioritized and actionable recommendations

Next Steps

:Create a practical action plan divided into

- Immediate actions \*
- Short-term actions \*
- Medium-term actions \*

---

## Recommendations Framework .17

:Every major recommendation must contain

Finding → Evidence → Business Impact → Recommended Action → Priority

:Prioritize recommendations as

- Critical \*
- High \*
- Medium \*
- Low \*

.Avoid generic recommendations

.Recommendations must be directly connected to findings from the dataset

---

## Technical & Excel Requirements .18

:Use the appropriate tools available in the workbook environment, including

- PivotTables \*
- PivotCharts \*
- Excel formulas \*
- Power Query \*
- Calculated columns \*
- Conditional formatting \*
- Data validation \*
- KPI calculations \*
- Time intelligence calculations \*
- Segmentation logic \*

.Use professional formatting

:Apply

- Clear sheet names \*
- Consistent headers \*



Appropriate number formats \*  
Percentages \*  
Currency formatting \*  
Conditional formatting \*  
Professional dashboard layout \*  
Clear titles \*  
Clear business labels \*  
Readable charts \*

.Do not modify the original raw data

.All calculations and outputs should be created in new worksheets

---

## Evidence & Insight Rules .19

:These rules are mandatory

.Every important insight must be supported by actual data .1

.Never invent values .2

.Never guess missing information .3

.Never assume causation from correlation .4

.Clearly identify limitations .5

:If a metric cannot be calculated because required columns are missing, explicitly state .6

”.Not available due to missing required data“

.If a conclusion is uncertain, clearly state the limitation .7

Do not remove data simply because it looks unusual. First determine whether it is an .8

.error, valid behavior, or an anomaly

.Preserve the original dataset .9

.Use calculated evidence wherever possible .10

---

## Final Objective

Transform the workbook from a collection of raw marketing, sales, and customer data into a  
.complete Marketing & Sales Intelligence System

:The final output should allow management to quickly understand

?What happened

?Why did it happen

?Which campaigns and channels caused it

?Which customers are driving the business

?Where are we losing customers or conversions

?What products and regions are performing  
?Where are the biggest risks  
?Where are the biggest opportunities  
?What should the business do next

The final analysis must be data-driven, evidence-based, commercially relevant, visually  
.clear, and presentation-ready

## Role-2

Act as a Senior Marketing Data Analyst, Business Intelligence Consultant, Data Storytelling  
.Expert, and Power BI Dashboard Architect

I will provide you with a completed Jupyter/Kaggle Notebook containing the full analysis of a  
Marketing & Customer dataset, including data cleaning, preprocessing, exploratory data  
.analysis (EDA), statistical analysis, derived features, and initial insights

.Your task is NOT simply to summarize the notebook

Your task is to study the entire notebook deeply and transform the existing analysis into a  
.professional, executive-level, business-focused Power BI dashboard strategy

The final dashboard must look like a real-world Marketing & Customer Intelligence  
Dashboard that could be presented to a Marketing Manager, Growth Manager, CRM  
.Manager, or executive team

---

## FIRST: STUDY THE ENTIRE NOTEBOOK .1

:Before suggesting any dashboard

.Read the entire notebook from beginning to end .1

:Understand .2

Dataset structure \*

Number of rows and columns \*

Every column and its business meaning \*

Data types \*

Missing values \*

Duplicates \*

Outliers \*

Categorical variables \*

Numerical variables \*

- Date variables \*
- Customer-related variables \*
- Marketing campaign variables \*
- Engagement variables \*
- Purchase/revenue variables \*
- Derived/calculated columns \*
- .Review all cleaning decisions already performed .3
- .Review every EDA result .4
- .Review every relationship discovered between variables .5
- .Identify statistically or commercially meaningful patterns .6
- Do NOT modify or contradict the existing cleaned dataset unless you explicitly identify a .7
- .data-quality issue
- .Use evidence from the notebook and dataset only .8
- .Do not invent statistics, percentages, relationships, or business conclusions .9

---

## THINK LIKE A BUSINESS ANALYST .2

- .Do not create a dashboard simply because a variable exists
- .Every KPI, chart, card, filter, and visual must answer a meaningful business question

:For every proposed visual, ask

"?What business decision would this visual help someone make"

.If the answer is unclear, DO NOT recommend the visual

:Prioritize insights related to

- Customer acquisition \*
- Customer engagement \*
- Marketing campaign performance \*
- Conversion \*
- Customer segmentation \*
- Revenue \*
- Customer value \*
- Retention / repeat purchasing \*
- Channel effectiveness \*
- Customer behavior \*
- Product/category performance \*
- Marketing efficiency \*
- Customer lifecycle \*
- High-value vs low-value customers \*

### IDENTIFY THE STRONGEST BUSINESS STORY .3

After studying the notebook, determine the 3–5 strongest business stories hidden in the .data

.Do not choose stories simply because they are easy to visualize

:Look for unexpected or high-impact findings such as

- A channel generating high engagement but low conversion \*
- A campaign generating many responses but weak revenue \*
- A small customer segment generating a disproportionately large share of revenue \*
- Customers with specific behavioral characteristics having significantly higher purchase value \*
- High-income customers not necessarily being the most valuable \*
- Discount usage being associated with different purchasing behavior \*
- Website/email/social engagement translating differently into purchases \*
- Certain customer segments responding better to certain campaigns \*
- Strong acquisition but weak retention \*
- High campaign response but poor downstream conversion \*
- Customers who engage frequently but spend less \*
- Customers with low engagement but unusually high revenue \*
- Age/income/education/marital status or other demographic variables showing meaningful behavioral differences \*
- Product preferences associated with customer value \*
- Purchase frequency vs monetary value patterns \*
- Recency/frequency/monetary relationships if available \*
- Any other counterintuitive relationship supported by the data \*

:For each business story, explain

- ?What is happening .1
- ?How strong is the evidence .2
- ?Why does it matter .3
- ?What business decision could be made .4
- ?Which dashboard visual should communicate it .5

\_\_\_\_\_

### DESIGN A 3-PAGE DASHBOARD .4

.Create a 3-page dashboard architecture

.Do not make the pages repetitive

.Each page must have a distinct business purpose

:Recommended structure

## PAGE 1 — EXECUTIVE MARKETING OVERVIEW

:Purpose

Give a manager a complete understanding of the overall health of customers, marketing activity, engagement, and revenue within 10–15 seconds

:The page should answer

- ?How are we performing overall \*
- ?How many customers do we have \*
- ?How much revenue/value are we generating \*
- ?How engaged are customers \*
- ?How effective are our marketing campaigns \*
- ?What are the most important trends \*
- ?What should management pay attention to \*

## PAGE 1 KPI CARDS

.Determine the most important KPIs from the actual dataset

:Potential KPIs include

- Total Customers \*
- Total Revenue / Total Spend \*
- Average Customer Value \*
- Average Order Value if available \*
- Purchase Rate / Conversion Rate \*
- Campaign Response Rate \*
- Average Purchase Frequency \*
- Repeat Customer Rate \*
- Average Customer Income \*
- Marketing Engagement Rate \*
- Campaign Acceptance Rate \*
- Website Engagement \*
- Email Engagement \*
- Customer Retention / Repeat Purchase Rate if measurable \*

.Do NOT blindly use all of these

.Select approximately 5–7 strongest KPIs based on the actual dataset

:For EACH KPI provide

- KPI name \*
- Exact business definition \*
- Exact formula \*

Required columns \*  
Why it matters \*  
Recommended Power BI DAX measure \*  
:Whether it should be displayed as \*  
Number \*  
Currency \*  
Percentage \*  
Average \*  
Ratio \*  
Recommended icon \*  
Recommended comparison/benchmark \*  
Recommended tooltip \*

:Example format

:KPI  
Customer Conversion Rate

:Definition  
.Percentage of customers who made at least one purchase

:Formula  
$$\frac{\text{Purchasing Customers}}{\text{Total Customers}}$$

:Business value  
.Measures how effectively customer engagement translates into actual purchasing behavior

:DAX  
.Provide the exact DAX measure based on the actual column names

---

## PAGE 1 VISUAL LAYOUT .5

.Design the page from top to bottom and left to right

:Tell me EXACTLY

Header

Dashboard title \*  
Subtitle \*  
Date/filter context \*  
Recommended logo/icon if useful \*

KPI Row

.Specify the exact order of KPI cards

:Example

Total Customers] [Revenue] [Conversion Rate] [Avg Customer Value] [Campaign Response]  
[Rate]

.But determine the final KPIs from the actual data

Main Visual Area

.Recommend approximately 4–6 visuals

:For every visual provide

- Exact chart type \*
- Exact title \*
- X-axis \*
- Y-axis \*
- Legend \*
- Values \*
- Aggregation \*
- Sorting \*
- Filters \*
- Drill-down if useful \*
- Tooltip \*
- Why this visual is important \*
- What insight it should reveal \*

:Possible visual types

- Line chart \*
- Clustered bar chart \*
- Stacked bar chart \*
- Donut chart only when genuinely appropriate \*
- Treemap \*
- Scatter plot \*
- Matrix \*
- KPI card \*
- Decomposition tree \*
- Funnel \*
- Waterfall \*
- Ribbon chart \*
- Map only if geographic data actually exists \*
- Small multiples if useful \*

.Avoid unnecessary pie/donut charts

\_\_\_\_\_

## PAGE 2 — CUSTOMER SEGMENTATION & CUSTOMER VALUE .6

:Create a dedicated page focused on

”?Who are our most valuable customers, and what differentiates them“

.This page should investigate customer behavior and segmentation

:Possible dimensions

- Age \*
- Income \*
- Education \*
- Marital status \*
- Household composition \*
- Children/teenagers \*
- Customer tenure \*
- Purchase frequency \*
- Total spending \*
- Product preferences \*
- Discount behavior \*
- Web visits \*
- Web purchases \*
- Catalog purchases \*
- Store purchases \*
- Deal purchases \*
- Campaign responses \*

.Use ONLY variables actually present in the dataset

\_\_\_\_\_

## PAGE 2 REQUIRED ANALYSIS

.Identify meaningful customer segments

.Do not create arbitrary segments just for visualization

:Possible approaches

A. Revenue / Spending Segments

:For example

- High Value \*
- Medium Value \*
- Low Value \*



.Only recommend this if justified by the distribution

## B. Engagement Segments

:For example

Highly Engaged \*  
Moderately Engaged \*  
Low Engagement \*

.Only if supported by the data

## C. Behavioral Segments

:Identify meaningful combinations such as

High engagement + high spending \*  
High engagement + low spending \*  
Low engagement + high spending \*  
Low engagement + low spending \*

.This is especially important because it can reveal hidden customer opportunities

\_\_\_\_\_

## PAGE 2 VISUALS .7

.Recommend exact visuals

:Strong candidates may include

Scatter Plot

:Example

:X-axis  
Customer Engagement

:Y-axis  
Total Spending

:Bubble size  
Purchase Frequency / Customer Value

:Color  
Customer Segment

:Purpose

Identify customers/segments with high engagement but unexpectedly low spending and vice versa

Bar Chart

:Top customer segments by

Revenue \*  
Customer count \*  
Average spend \*  
Conversion \*

Box Plot Alternative

.If Power BI implementation requires a custom visual, explain whether it is worth using

Heatmap

:Possible relationship

Customer Segment × Product Category

:Metric  
Average Spending / Purchase Rate

Demographic Analysis

:Only if meaningful

Age Group vs Spending

Income Group vs Spending

Education vs Spending

Household Profile vs Spending

But do not fill the dashboard with demographic charts unless the notebook proves they matter

\_\_\_\_\_

:Create a dedicated page answering

”?Which marketing activities actually generate customer action and value“

:Focus on

- Campaign responses \*
- Campaign acceptance \*
- Channel performance \*
- Email \*
- Web \*
- Catalog \*
- Store \*
- Social media if available \*
- Marketing engagement \*
- Conversion \*
- Revenue contribution \*
- Customer response behavior \*

---

## PAGE 3 — MARKETING FUNNEL .9

:If the data supports it, create a marketing funnel such as

Reach / Exposure  
↓  
Engagement  
↓  
Response  
↓  
Purchase  
↓  
Repeat Purchase

.However, DO NOT assume these stages exist

.First verify which stages can actually be measured from the dataset

:For every funnel stage provide

- (Column(s) \*
- Definition \*
- Formula \*
- Conversion rate \*
- Drop-off rate \*
- Business interpretation \*

:Most importantly

.Identify the largest drop-off point

.This should become one of the major insights in the dashboard

---

## CAMPAIGN PERFORMANCE ANALYSIS .10

:Determine which campaigns are

- Best at generating responses \*
- Best at generating purchases \*
- Best at generating revenue \*
- Best at engaging high-value customers \*
- Weak despite high engagement \*
- Strong in volume but weak in quality \*

.Create a performance matrix if appropriate

:Example

Campaign | Reach | Engagement | Response | Conversion | Revenue | Customer Value

:Explain which campaigns should

- Scale \*
- Optimize \*
- Reduce \*
- Retarget \*

.Do not make these recommendations unless the data supports them

---

## CHANNEL ANALYSIS .11

.Compare available channels

:For each channel calculate, where possible

- Engagement \*
- Customers reached \*
- Response \*
- Conversion \*
- Revenue \*
- Average customer value \*
- Purchase frequency \*

:Then identify

High Volume / High Value

Best-performing channels

High Volume / Low Value

Potential efficiency problem

Low Volume / High Value

Potential scaling opportunity

Low Volume / Low Value

Potentially weak channel

.This 2×2 logic should be considered for a scatter plot

---

## CREATIVE / ADVANCED INSIGHTS .12

I specifically want you to search the notebook/data for non-obvious insights, not generic  
:statements such as

”.Most customers are X“

:Prioritize insights like

Insight Type 1 — Mismatch

.High engagement but low conversion

Insight Type 2 — Hidden Value

.Small segment but disproportionately high revenue

Insight Type 3 — Opportunity

.Low-performing segment that shows strong engagement and could be converted

Insight Type 4 — Efficiency

.A campaign/channel generates many responses but weak customer value

Insight Type 5 — Behavioral Pattern

.A specific behavior is strongly associated with high spending

Insight Type 6 — Risk

.A large customer group shows low engagement or low repeat purchasing

Insight Type 7 — Optimization

.Certain customer segments perform better under certain campaign/channel strategies

Insight Type 8 — Contradiction

.A commonly assumed relationship is NOT supported by the data

.These insights are much more valuable than simple descriptive statistics

\_\_\_\_\_

KPI HIERARCHY .13

:Create a hierarchy of KPIs

Executive KPIs

.metrics only 7–5

Diagnostic KPIs

.Metrics used to explain why the executive KPI is changing

Supporting Metrics

.Metrics used in tooltips, drill-throughs, and detailed analysis

.Show the relationship between them

:For example

Revenue

↓

Number of Purchasing Customers

+

Average Customer Value

↓

Purchase Frequency

+

Average Spend

:Or

Campaign Success  
↓  
Response Rate  
↓  
Conversion Rate  
↓  
Revenue per Customer

.Build the KPI tree based on the actual data

---

## FILTERS / SLICERS .14

.Recommend the exact slicers that should appear

:Possible slicers

Age Group \*  
Education \*  
Marital Status \*  
Income Group \*  
Customer Segment \*  
Campaign \*  
Channel \*  
Product Category \*  
Date / Year / Month \*  
Customer Type \*

.But ONLY recommend dimensions that are genuinely useful

:Specify

Which slicers belong on Page 1 \*  
Which belong on Page 2 \*  
Which belong on Page 3 \*  
Which should be synced across pages \*  
Which should remain page-specific \*

.Avoid slicer overload

---

## CROSS-PAGE INTERACTIONS .15

.Design the interaction between the 3 pages

:Explain

Page 1 → Page 2

?If the user clicks a segment, what should happen

Page 1 → Page 3

?If the user clicks a campaign/channel, what should happen

Page 2 → Page 3

?If a high-value segment is selected, how should campaign performance change

:Recommend

Cross-filtering \*

Drill-through \*

Tooltip pages \*

Bookmarks \*

Buttons \*

Navigation \*

Reset Filters button \*

.Explain exactly where each should be used

\_\_\_\_\_

## DRILL-THROUGH PAGES .16

.Determine whether a drill-through page is worth creating

:Possible drill-through

Customer Segment Detail

:When a user clicks a customer segment

:Show

Customer count \*

Revenue \*

Average spend \*

Purchase frequency \*

Engagement \*

Campaign response \*



Product preference \*  
Channel behavior \*

.If drill-through is not necessary, explicitly say so

\_\_\_\_\_

## TOOLTIP DESIGN .17

.For each major visual, design a professional tooltip

:Example

:When hovering over a Campaign

Campaign Name  
Customers Reached  
Response Rate  
Conversion Rate  
Revenue  
Average Customer Value  
vs Overall Average

:Do this for

Customer Segment \*  
Campaign \*  
Channel \*  
Product Category \*  
Time Period \*

\_\_\_\_\_

## DAX REQUIREMENTS .18

Based on the actual column names in the notebook, provide the important Power BI DAX  
.measures

.Do NOT give generic placeholders

:For each measure provide

Measure name .1  
Exact DAX .2  
Business definition .3  
Where it will be used .4

:Include measures for

Total Customers \*  
Purchasing Customers \*  
Total Revenue / Total Spending \*  
Average Customer Value \*  
Conversion Rate \*  
Response Rate \*  
Repeat Customer Rate if possible \*  
Average Purchase Frequency \*  
Channel metrics \*  
Campaign metrics \*  
Segment metrics \*  
Any advanced KPIs you recommend \*

:Also identify which calculations should be

Measures \*  
Calculated columns \*  
Power Query transformations \*

.Avoid creating calculated columns when a measure is more appropriate

\_\_\_\_\_

## CHART SELECTION RULES .19

.For every chart explain WHY you selected it

:Use

Bar Chart

.For comparing categories

Line Chart

.For trends over time

Scatter Plot

.For relationships between two metrics and identifying outliers/opportunities

Matrix

.For detailed comparisons

Funnel

.Only when sequential stages truly exist

Waterfall

.For explaining contribution to change

Treemap

.For hierarchical contribution such as revenue by category

Decomposition Tree

.For interactive root-cause analysis

Donut

.Only for simple composition with a small number of categories

.Do not recommend charts just because they look attractive

---

## VISUAL DESIGN SYSTEM .20

.Create a professional visual design recommendation

:Specify

Theme

:Recommend a modern

Corporate \*

Clean \*

Premium \*

Data-driven \*

.style

Colors

:Provide exact HEX codes for

Primary \*

Secondary \*

Positive \*

Negative \*

Neutral \*

Background \*  
Card background \*  
Text \*  
Gridlines \*

.Use a restrained palette

.Do not use excessive colors

:Use consistent semantic colors

Green = positive  
Red = negative  
Gray = neutral  
Accent = selected/highlighted data

\_\_\_\_\_

## ICONS .21

.For every KPI recommend an appropriate icon

:Examples

Total Customers → People / Users icon

Revenue → Dollar / Coins icon

Conversion → Target / Bullseye icon

Engagement → Activity / Pulse icon

Campaigns → Megaphone icon

Customer Value → Crown / Wallet icon

Retention → Repeat / Refresh icon

Purchases → Shopping Cart icon

.But choose the final icons based on the actual KPIs

:Also suggest where to get the icons

Power BI built-in icons \*  
Microsoft Fluent UI Icons \*  
Font Awesome \*  
Material Symbols \*

.Keep icons consistent

\_\_\_\_\_

## EXACT PAGE WIREFRAME .22

.For EACH PAGE provide an ASCII wireframe

:Example

\_\_\_\_\_

|  |                                               |             |  |
|--|-----------------------------------------------|-------------|--|
|  | Dashboard Title                               | Date Filter |  |
|  | KPI 1   KPI 2   KPI 3   KPI 4   KPI 5   KPI 6 |             |  |
|  |                                               |             |  |
|  | Main Trend / Performance Visual               |             |  |
|  |                                               |             |  |
|  | Segment Performance   Channel Performance     |             |  |
|  |                                               |             |  |
|  | Key Insight / Recommendation                  |             |  |

.Then replace the generic labels with the actual visuals you recommend

\_\_\_\_\_

## INSIGHT CARDS .23

.Each page should contain 1–3 dynamic insight areas

.Do NOT simply write static observations

:Recommend insight statements such as

”.High-value customers represent X% of customers but generate Y% of revenue“

:or

”.Channel A has higher engagement than Channel B, but lower conversion“

.However, only generate actual numbers after analyzing the dataset

:For each insight

Give the exact finding \*  
Give the supporting metric \*  
Explain why it matters \*  
Give a business recommendation \*

---

## EXECUTIVE RECOMMENDATIONS .24

:At the end, provide

Top 5 Business Insights

.Rank them from most important to least important

:For each

Finding  
Evidence  
Business Impact  
Recommended Action

---

## DASHBOARD STORYLINE .25

:The dashboard must tell a story

Page 1

”?What is happening“

↓

Page 2

”?Who is driving the results“

↓

Page 3

”?Why is it happening and what should marketing do“

.Make sure the three pages logically connect

---

## FINAL DELIVERABLE .26

:At the end, produce a complete implementation blueprint with

- Dashboard name .1
  - Business objective .2
  - Target audience .3
  - Three page names .4
  - Purpose of each page .5
  - KPI list .6
  - Exact KPI formulas .7
  - DAX measures .8
  - Exact visuals .9
  - Chart types .10
  - Axes .11
  - Legends .12
  - Filters .13
  - Slicers .14
  - Drill-through .15
  - Tooltips .16
  - Interactions .17
  - Icons .18
  - Color palette .19
  - Page layouts .20
  - Business insights .21
  - Business recommendations .22
  - Data model recommendations .23
  - Any additional calculated columns/measures needed .24
  - Recommended dashboard title and subtitle .25
- 

## IMPORTANT QUALITY RULES .27

:Follow these rules strictly

- .Do not invent data \*
- .Do not invent KPIs \*
- .Do not invent relationships \*
- .Do not create meaningless visuals \*
- .Do not repeat the same metric across multiple visuals unless there is a clear reason \*
- .Do not overload the dashboard \*
- .Prefer fewer, stronger visuals \*
- .Every visual must answer a business question \*
- .Every KPI must have a business purpose \*
- .Every page must have a clear narrative \*
- .Prioritize actionable insights over descriptive statistics \*
- .Use the actual column names from the notebook \*

- .Use exact formulas based on the actual dataset \*
- .If a metric cannot be calculated reliably, explicitly say so \*
- If a suggested analysis is not supported by the available data, explicitly reject it and explain \*
- .why
- .Distinguish between correlation and causation \*
- .Do not claim that one variable “causes” another unless the analysis supports causality \*
- .Highlight data limitations when relevant \*

---

## MOST IMPORTANT REQUIREMENT

- .Do not give me a generic dashboard template
- .I want you to discover the dashboard from the data
- .The notebook already contains the analysis
- :Your job is to determine
- ?What are the most valuable things a marketing manager should know from this dataset
- .Then design the Power BI dashboard around those discoveries
- I want the final result to be detailed enough that I can open Power BI immediately and build the dashboard visual by visual, KPI by KPI, page by page, without having to decide the structure myself